Pei-Hsin Chiu

Iowa City, IA | Open to Relocation | skuldchiu@gmail.com | LinkedIn| Portfolio
Ph.D. Candidate | Data Science & Machine Learning | Healthcare AI

PROFESSIONAL SUMMARY

Ph.D. candidate in Electrical and Computer Engineering with experience in data science, machine learning, deep learning, and statistical analysis. Skilled in Python, PyTorch, predictive modeling, model evaluation, and analyzing complex medical imaging and structured clinical datasets. Research experience spans interpretable representation learning, longitudinal forecasting, EHR-based clinical prediction, and multimodal modeling. Published researcher with experience translating clinical and scientific problems into quantitative machine learning workflows.

TECHNICAL SKILLS

Programming & Tools: Python, SQL, Git, GitHub, MATLAB, NumPy, pandas
Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, VAEs, regression, classification, representation learning
Data Science & Statistical Analysis: Data preprocessing, feature engineering, statistical analysis, predictive modeling, cross-validation, model evaluation, time-series forecasting
Healthcare AI & Medical Imaging: EHR-based clinical prediction, OCT, ultrasound imaging, 3D CT, medical image segmentation, nnU-Net, SimpleITK
NLP & Large Language Models: BERT, DistilBERT, RoBERTa, GPT-2, Hugging Face, LoRA, prompt-based model evaluation
Scientific Computing: Molecular dynamics simulation, simulation data analysis, computational modeling

PROFESSIONAL EXPERIENCE

University of Iowa, Electrical and Computer Engineering
Research Assistant | Aug 2022 – Present

- Designed and validated an attribute-guided variational autoencoder for interpretable representation learning from 4,900+ glaucoma OCT scans, learning anatomically meaningful latent representations associated with clinical biomarkers (R^2 up to 0.95) while maintaining high reconstruction quality (SSIM ~0.73).
- Developed longitudinal deep learning models for time-series forecasting of glaucoma progression from OCT and visual field data collected at irregular clinical follow-up intervals.
- Developed multimodal models integrating OCT imaging with visual field measurements to study patient-level structure-function relationships in glaucoma.
- Developed and evaluated 3D CT-based medical image segmentation models using nnU-Net for the ascending aorta and aortic root sinuses, including image resampling, expert-annotated mask preparation, cross-validation, Dice evaluation, and qualitative error analysis.
- Collaborated with ophthalmology, cardiac surgery, and engineering teams to translate clinical research questions into machine learning workflows and evaluation strategies.

National Taiwan University, Biomedical Electronics and Bioinformatics
Research Assistant | May 2021 – May 2022

- Developed and trained machine learning models for pediatric respiratory pathogen prediction and ICU mortality risk assessment, working with EHR data from a clinical cohort of 12,000+ admissions.
- Performed clinical data cleaning, preprocessing, feature engineering, and statistical analysis using demographics, vital signs, laboratory measurements, and underlying conditions.
- Evaluated and visualized predictive model performance using AUROC, AUPRC, and interpretable feature analysis, contributing to two peer-reviewed publications in pediatric infectious disease and critical care.

Institute for Information Industry
Machine Learning Development Intern | Feb 2020 – Aug 2020

- Supported machine learning prototyping for a 3D lung medical imaging project, gaining exposure to medical imaging workflows and model development.
- Implemented Python-based data processing and machine learning code using NumPy and deep learning frameworks.

EDUCATION

University of Iowa, Ph.D. in Electrical and Computer Engineering Aug 2022 – Expected May 2027
Dissertation Research: Interpretable deep learning for OCT-based glaucoma analysis and longitudinal disease progression modeling

National Taiwan University, M.S. in Civil Engineering Sep 2018 – Jun 2020
Thesis: Diagnosis of Supraspinatus Calcific Tendinopathy using Convolutional Neural Networks

National Taiwan University, B.S. in Civil Engineering Sep 2014 – Jun 2018

SELECTED RESEARCH PROJECTS

Large Language Models – Disaster Tweet Classification
University of Iowa | ECE:5995 Large Language Models, Spring 2025

- Fine-tuned BERT, DistilBERT, and RoBERTa using pretrained Hugging Face checkpoints for disaster-tweet classification and implemented parameter-efficient fine-tuning with LoRA, comparing performance against classical NLP baselines.
- Evaluated GPT-2 using zero-shot, few-shot, and Chain-of-Thought prompting, analyzing accuracy, precision, recall, F1 score, and model failure cases.

PUBLICATIONS

- **Pei-Hsin Chiu**, B. A. Johnson, E. F. Linton, A. E. Pouw, M. Wall, Y. H. Kwon, R. H. Kardon, J.-K. Wang, and M. K. Garvin. "abVAE: Attribute-Based Booster Variational Autoencoder for Interpretable Latent Presentation in Optical Coherence Tomography of Glaucomatous Eyes." *Ophthalmic Medical Image Analysis (OMIA 2025), Lecture Notes in Computer Science*, vol. 16209, pp. 137–146, 2026. doi:10.1007/978-3-032-10351-2_14.
- **Pei-Hsin Chiu**, M. Boudier-Revéret, S.-W. Chang, C.-H. Wu, W.-S. Chen, L. Özçakar. "Deep Learning for Detecting Supraspinatus Calcific Tendinopathy on Ultrasound Images." *Journal of Medical Ultrasound*. 2022;30(3):196–202. doi:10.4103/jmu.jmu_182_21.
- S.-R. Lin, J.-H. Wu, Y.-C. Liu, **Pei-Hsin Chiu**, T.-H. Chang, E.-T. Wu, C.-C. Chou, L.-Y. Chang, and F.-P. Lai. "Machine learning models to evaluate mortality in pediatric patients with pneumonia in the intensive care unit." Pediatric Pulmonology, 59(5), 1256–1265, 2024. doi:10.1002/ppul.26897.
- T.-H. Chang, Y.-C. Liu, S.-R. Lin, **Pei-Hsin Chiu**, C.-C. Chou, L.-Y. Chang, and F.-P. Lai. "Clinical characteristics of hospitalized children with community-acquired pneumonia and respiratory infections: Using machine learning approaches to support pathogen prediction at admission." *Journal of Microbiology, Immunology and Infection*, 56(4), 772–781, 2023. doi:10.1016/j.jmii.2023.04.011.
- W.-H. Hui*, **Pei-Hsin Chiu***, I.-I. Ng, S.-W. Chang, C.-C. Chou, and H.-H. Chen. "Unraveling the molecular mechanism of collagen flexibility during physiological warmup using molecular dynamics simulation and machine learning." Computational and Structural Biotechnology Journal, 21, 1630–1638, 2023. doi:10.1016/j.csbj.2023.02.017.
- Y.-L. Chen, Y. Chiang, **Pei-Hsin Chiu**, I.-C. Huang, Y.-B. Xiao, S.-W. Chang, and C.-W. Huang. "High-Dimensional Phase Space Reconstruction with a Convolutional Neural Network for Structural Health Monitoring." Sensors, 21(10), 3514, 2021. doi:10.3390/s21103514.

* Sharing equal contribution

HONORS & AWARDS

Graduate Student Poster Award, Electrical and Computer Engineering, University of Iowa College of Engineering, 2026
Second Paper Award, Taiwan Society of Ultrasound in Medicine, 2023
Best Presentation Award, Singapore Healthcare AI Datathon and Expo, 2021